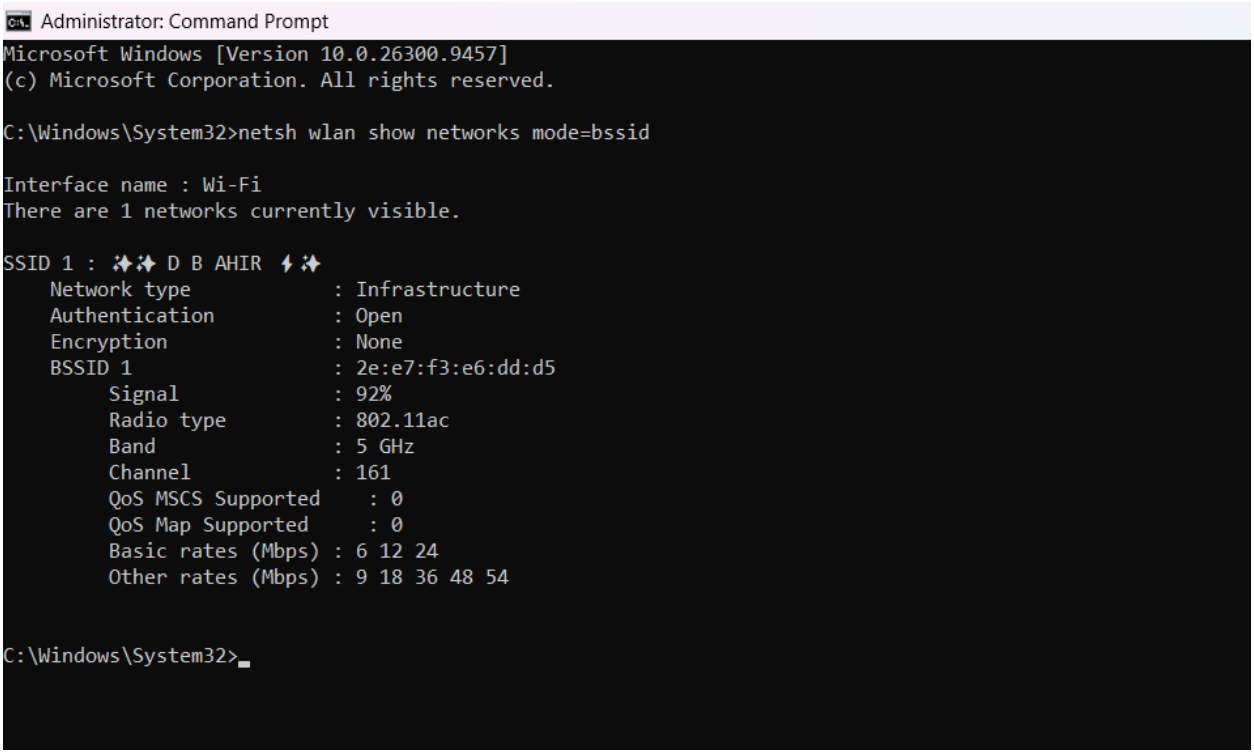
Wireless Technology

Tasks :-

1.Scan for available Wi-Fi networks on your laptop or phone and list the SSIDs you find, noting which ones are open (unsecured) and which require a password.

Ans :-



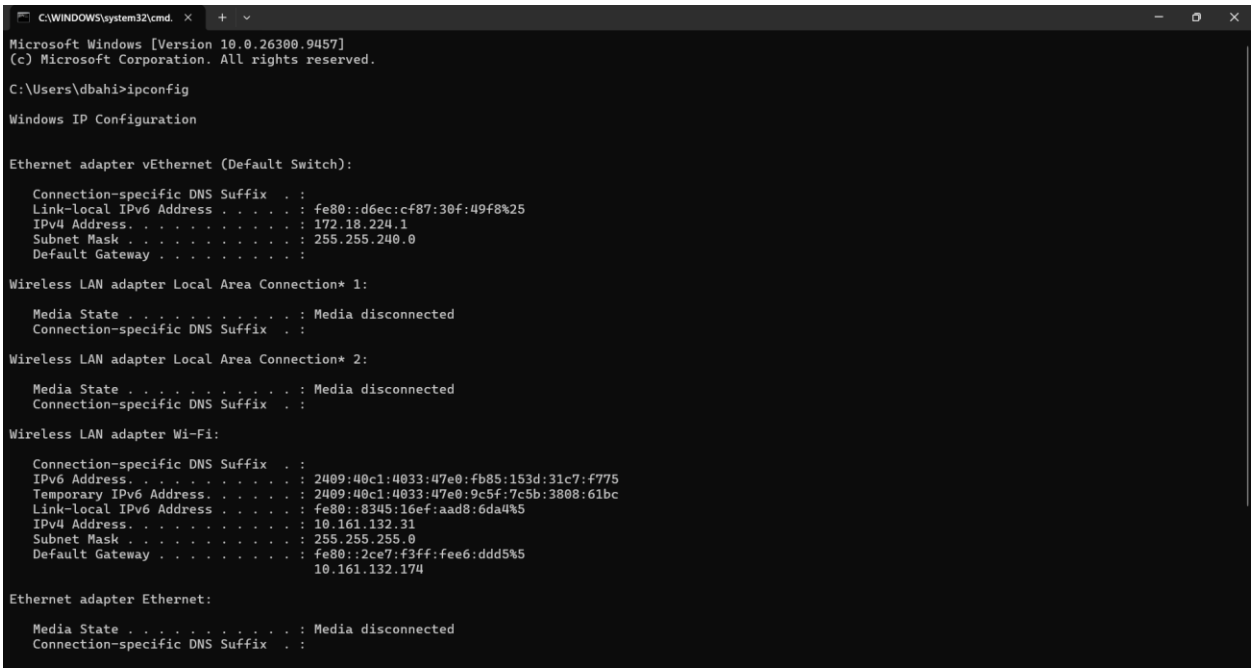
- **Result :-**

The Wi-Fi scanning process was successfully completed using the Windows Command Prompt. One wireless network was detected. The network uses Open authentication with no encryption, indicating that it does not require a Wi-Fi password to connect.

**2.Set up a mobile hotspot on your phone, connect a laptop to it, and check what IP address your laptop receives from the hotspot.

Hint: On Windows, use 'ipconfig' in Command Prompt; on Mac/Linux, use 'ifconfig' in Terminal.**

Ans :-



3.Research and explain in your own words what WPA2 and WPA3 are, and list two differences between them relevant to security.

Ans :-

1. What is WPA2?

WPA2 (Wi-Fi Protected Access 2) is a wireless network security protocol used to protect Wi-Fi connections. It uses encryption to help prevent unauthorized users from reading data transmitted over a wireless network.

WPA2 commonly uses AES-based CCMP encryption to protect wireless communication.

2. What is WPA3?

WPA3 (Wi-Fi Protected Access 3) is a newer generation of Wi-Fi security protocol developed to improve the protection provided by WPA2.

It introduces stronger protection for password-based authentication, better resistance to password-guessing attacks, and improved security for certain open Wi-Fi networks.

- **Differences Between WPA2 and WPA3 :-**

Comparison	WPA2	WPA3
1. Authentication	Uses Pre-Shared Key (PSK) in Personal mode, allowing captured handshakes to be subjected to offline password guessing.	Uses SAE (Simultaneous Authentication of Equals), which protects against offline password-guessing attacks.
2. Management Frame Protection	Protection is optional.	Protected Management Frames (PMF) are mandatory, helping prevent forged deauthentication attacks.

4.Find a news article or real-world case from the past 3 years where a public Wi-Fi network was used in a cyber attack, and summarize how the attack happened and what could have prevented it.

Ans :-

- **Public Wi-Fi Cyber Attack Case Study: The Australian "Evil Twin" Airport Attacks (2024–2025)**
  - **The Incident :-**

In a significant case that concluded in late 2025, a 44-year-old Australian man named Michael Clapsis was sentenced to 7 years and 4 months in prison for running a sophisticated "evil twin" Wi-Fi attack campaign at airports and on domestic flights across Australia. The attacks took place at major airports in Perth, Melbourne, and Adelaide, as well as on certain domestic flights, spanning from as early as 2015 until his arrest in April 2024**.**

  - **How the Attack Happened**

The attacker used a portable wireless hacking device called a "Wi-Fi Pineapple" — a tool commonly used for penetration testing but repurposed here for malicious credential theft.

- **The attack unfolded in several stages:**

1. Rogue Access Point Creation

2. Captive Portal Phishing

3. Credential Harvesting

4. Anti-Forensics and Obstruction

- **What Could Have Prevented It**

For Users:

1. Disable automatic Wi-Fi connections in public places. This prevents devices from silently connecting to rogue access points that mimic known networks.
2. Always use a reputable VPN when connecting to public Wi-Fi.
3. Treat captive portals with skepticism. Any public Wi-Fi login page requesting personal account credentials (email, social media, banking passwords) should be considered highly suspicious.
4. Avoid sensitive activities like online banking or accessing private accounts while on public Wi-Fi.